
1. Apartments

Program Name: Apartments.java

Input File: apartments.dat

Judges Input File

```
5
8
3 5
2 4
7 0
5 3
8 3
9 5
10 3
14 0
10
101 5
102 4
103 5
104 3
105 5
201 0
202 2
203 5
204 6
205 3
5
1 0
2 0
3 0
4 0
5 0
7
1 4
2 4
3 4
5 4
6 4
7 4
8 4
8
10 5
11 5
12 5
13 3
14 3
15 3
16 0
17 5
```

Judges Output to Screen

```
YES 2
NO 1
YES 5
YES 0
NO 1
```

2. Blob Top

Program Name: BlobTop.java **Input File: blobtop.dat**

Input File: blobtop.dat

Judges Input File

```

4
7 8 4
      *
****
.***
****
      **
      *
    ****
    *
  **
4 1
5 3
7 8
1 5
5 6 4
    ***
****
    *
    **
  *
.***
**
. **
5 6
1 6
2 1
5 1
3 8 3
***
    **
***
  *
***
.
2 2
3 3
2 5
9 13 7
*****
*****
*****
** *
**
*
*
.
.
****
*****
*****
4 3
7 8
5 13
7 2
2 7
7 12
9 4

```

2. Blob Top (cont.)

Judges Output to Screen

```
2 1
NOT A BLOB
NOT A BLOB
1 5
1 4
1 4
2 1
4 2
1 1
1 1
1 7
1 1
3 8
1 1
7 2
NOT A BLOB
7 2
7 2
```

3. Buying a Car

Program Name: Buying.java

Input File: buying.dat

Judges Input File

7

Honda Element--19--17950

Ford Edge--19--18130

Toyota RAV4--20--18885

Hyundai Santa Fe--17--19070

Subaru Outback--20--22100

Mazda 5--21--14980

Scion--22--16692

Judges Output to Screen

Honda Element \$13450 \$1952.63

Ford Edge \$13630 \$1952.63

Toyota RAV4 \$14385 \$2120.00

Hyundai Santa Fe \$14570 \$1558.82

Subaru Outback \$17600 \$2120.00

Mazda 5 \$10480 \$2271.43

Scion \$12192 \$2409.09

4. Cancer Cells

Program Name: Cancer.java

Input File: cancer.dat

Judges Input File

```
12
5
....@
##...
#....
...#.
##.##
3
.##
.#.
@##
3
##.
#..
@..
5
....@
##...
#....
...#.
##.##
3
...
.@.
...
5
#...#
...#.
#....
...##
..@..
5
#....
....
....
....
..@..
5
#...#
#.#.#
.#.#.
....#
.#@##
5
...##.
..#..
#....
#....
.#@..
```

(continued on next page)

4. Cancer Cells (cont.)

```
4
####
@###
####
####
```

```
5
.....
.....
..@..
.....
.....
5
@....
.....
.....
.....
.....#
```

Judges Output to Screen

```
3
10
-1
3
0
2
1
6
4
-1
0
1
```

5. CDs

Program Name: CDs.java Input File: cds.dat

Judges Input File

```
4
18
5 25
4 58
4 45
5 35
4 56
5 23
4 34
3 45
5 34
5 45
7 23
5 34
4 45
4 0
5 3
13 34
4 56
5 52
10
5 56
6 12
4 56
5 35
4 45
7 23
5 56
5 43
4 9
5 0
9
4 30
5 30
4 30
5 30
8 12
7 28
3 42
9 21
10 39
10
12 34
7 27
4 56
7 45
13 15
7 45
5 34
5 0
5 1
4 36
```

5. CDs (cont.)

Judges Output to Screen

PARTY #1: 7
PARTY #2: 4
PARTY #3: 3
PARTY #4: 6

6. De-Expansion

Program Name: Deexpansion.java

Input File: deexpansion.dat

Judges Input File

```
00101010000000000000101100000000001001011000000000011001110000000000000010000000
0001001011000000000011001110000000000000010000000000001011100000000000001011000
0000001010011000000000000000100000000000111011000000000010011100000000001001011
000000000011100110000000000100101100000000000111011000000000010000110000000000011
011000000000000000100000000000110011000000000100101100000000000110110000000001
0100110000000000011101000000000001011000000000000010100000000000010110000000000
0001010000000000000011100010000000000111011000000000011101100000000000100110000
0000000000010000000000000101110000000000001011000000000100101100000000001110110
000000000111001100000000000000010000000000100111100000000011101100000000010101
110000000000000001000000000001001110000000001010011000000000110001100000000011
11011000000000001101110000000001010011000000000100111000000000101001100000000
00010011000000000000000100000000001001011000000000010111000000000100001000000
0000101100000000000000010100000000000000101000000000111011000000000101011100
00000000111011000000000001001100000000000000100000000001100010000000000101100
000000000001010000000000000000010100000000010100110000000000100111000000000110
0011000000000101001100000000001110110000000000010111000000000000010000000000
1010010000000000010110000000000000010100000000000001000010000000001000011000000
000110001100000000001101011000000000110011100000000000110110000000001000011000
00000011001110000000000000101100000000000000100000000000011101000000000
```

(The actual input file contains only one line which appears to span many lines in the printed version.)

Judges Output to Screen

This is the original file.

Good thing you recovered it!

Pound #

Percent %

Backslash \

7. Selection Sort

Program Name: Selection.java

Input File: selection.dat

Judges Input File

```
7
45 15 12 -5 14 -3 8
12 -23 43 56 43 -1 -5
8 -2 -4 0 17 3 6 9 2 -2
8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8
3 4 2 5 1 6 8 7 0
12 9 0 -3 8 10
```

Judges Output to Screen

```
-5 15 12 45 14 -3 8
-5 -3 12 45 14 15 8
-5 -3 8 45 14 15 12
-5 -3 8 12 14 15 45
-5 -3 8 12 14 15 45
-5 -3 8 12 14 15 45

-23 12 43 56 43 -1 -5
-23 -5 43 56 43 -1 12
-23 -5 -1 56 43 43 12
-23 -5 -1 12 43 43 56
-23 -5 -1 12 43 43 56
-23 -5 -1 12 43 43 56

-4 -2 8 0 17 3 6 9 2 -2
-4 -2 8 0 17 3 6 9 2 -2
-4 -2 -2 0 17 3 6 9 2 8
-4 -2 -2 0 17 3 6 9 2 8
-4 -2 -2 0 2 3 6 9 17 8
-4 -2 -2 0 2 3 6 9 17 8
-4 -2 -2 0 2 3 6 9 17 8
-4 -2 -2 0 2 3 6 8 17 9
-4 -2 -2 0 2 3 6 8 9 17

1 7 6 5 4 3 2 8
1 2 6 5 4 3 7 8
1 2 3 5 4 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8

1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
1 2 3 4 5 6 7 8
```

(continued on next page)

7. Selection Sort (cont.)

```
0 4 2 5 1 6 8 7 3
0 1 2 5 4 6 8 7 3
0 1 2 5 4 6 8 7 3
0 1 2 3 4 6 8 7 5
0 1 2 3 4 6 8 7 5
0 1 2 3 4 5 8 7 6
0 1 2 3 4 5 6 7 8
0 1 2 3 4 5 6 7 8
```

```
-3 9 0 12 8 10
-3 0 9 12 8 10
-3 0 8 12 9 10
-3 0 8 9 12 10
-3 0 8 9 10 12
```

8. Smoothing an Image

Program Name: Smooth.java

Input File: smooth.dat

Judges Input File

```
10 5
 65 223 255 133 221 95 141 41 172 127
177 37 68 0 224 196 243 145 61 75
236 151 207 197 41 106 120 216 215 159
226 57 176 30 224 67 217 244 246 22
226 57 27 31 46 101 250 255 234 160
100 140 250 184 73 206 90 212 131 9
109 147 116 226 217 238 117 244 187 198
 24 19 86 162 5 227 189 1 41 21
 30 49 169 238 149 158 112 87 206 211
181 112 54 199 196 106 174 63 6 73
```

Judges Output to Screen

```
158 146 149 144 150 141 149 141 143 135
157 140 147 135 148 145 162 145 153 144
146 129 133 119 137 143 165 156 167 158
142 129 127 116 135 149 167 159 165 159
148 145 140 133 142 158 172 170 176 182
117 120 118 124 142 154 162 156 153 147
103 120 115 133 147 153 151 155 148 146
106 130 129 149 158 155 137 132 119 113
91 120 124 144 157 155 136 133 121 112
80 110 112 129 148 138 115 112 99 79
```

9. Superhero Day

Program Name: Superhero.java

Input File: superhero.dat

Judges Input File

```
*****
**$$$$$$$$$$$$$$$$**
** $$          $$ $ **
** $$          $$$  **
** $$$                **
*$$$$$$$$$          **
*$$$$$$$$$$$$$$$**
**$$$$$$$$$$$$$$$**
**$$$$$$$$$$$$$$$**
**          $$$$$$$$**
**          $$$**
**$$$          $$$**
*$$$$$$$$$**
**          **
**          **
***
*
```

Judges Output to Screen

```
*****
**$$$$$$$$$$$$$$$$**
** $ $$          $$ **
** $$$          $$ **
**          $$$ **
**          $$$$$$$$*
**          $$$$$$$$$$$$*
**$$$$$$$$$$$$$$$**
**$$$$$$$$$          **
**$$$                **
**$$          $$$**
**$$$          $$$**
**          **
**          **
**          **
***
*
```

10. Unwrapping a Spiral

Program Name: Unwrap.java

Input File: unwrap.dat

Judges Input File

```
10
1  2  3  4  5  6  7  8  9 10
11 12 13 14 15 16 17 18 19 20
21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48 49 50
51 52 53 54 55 56 57 58 59 60
61 62 63 64 65 66 67 68 69 70
71 72 73 74 75 76 77 78 79 80
81 82 83 84 85 86 87 88 89 90
91 92 93 94 95 96 97 98 99 100
```

Judges Output to Screen

```
1 2 3 4 5 6 7 8 9 10
20 30 40 50 60 70 80 90 100 99
98 97 96 95 94 93 92 91 81 71
61 51 41 31 21 11 12 13 14 15
16 17 18 19 29 39 49 59 69 79
89 88 87 86 85 84 83 82 72 62
52 42 32 22 23 24 25 26 27 28
38 48 58 68 78 77 76 75 74 73
63 53 43 33 34 35 36 37 47 57
67 66 65 64 54 44 45 46 56 55
```

11. Words

Program Name: Words.java

Input File: words.dat

Judges Input File

17

BEEBEE BEEFED BEEPED BEEPER BEETLE BEEVES BEEZER BENDEE BESEEM INWALL INWARD
BEWEEP BREEZE CHEESE CREESE DECREE DEEDED DEEMED DEEPEN DEEPER INWIND INWOVE
DEGREE DELETE EELIER EERIER EFFETE EKUELE ELEVEN EMCEED EMCEES INDRI INDUE
EMEERS EMERGE EMESES EMEUTE ENTRÉE EPEE GAGED GAGER GAGES GANGS INVOKE INWRAP
GAUGE GIGAS GIGHE GIGOT GIGUE GLOGG GLUGS GOGOS GOING GONGS INDOW INEPT INERT
GORGE GOUGE GREGO GRIGS GROGS GULAG GURGE HOGGS JAGGS JAGGY INFER INFIX INFOS
LEGGY LOGGY MIGGS MOGGY MUGGS MUGGY NAGGY NOGGS UREA UVEA ZOEIA INFRA INGLE
RAGGY SAGGY SOGGY VUGGS VUGGY WIGGY YEGGS AGOG EGGS EGGY GAGA INGOT INKER
GAGE GAGS GANG GIGA GIGS GLEG GLUG GOGO GONG GRIG GROG HOGG JAGG INK INN INS
MIGG MUGG NOGG VUGG YEGG EGG GAG GIG BEE CEE DEE EEL EKE EME ERE INTO INTRO
EVE EWE EYE FEE INBY INCH INFO INIA INKS INKY INLY INNS INRO OUTSWIM OUTVIES
INTI INTO INK INN INS EAU AA AE AI OE RAI A ROUE OUTSUNG OUTSWAM OUTSWUM
OUTTAKE OUTTALK OUTTASK OUTTELL OUTTOLD OUTTROT OUTTURN OUTVIED OUTVOTE
OUTYELL REWIDEN REWINDS REWIRED REWIRES REWOKEN REWORDS REWORKS REWOUND
REWOVEN REWRAPS REWRAPT REWRITE REWROTE REYNARD REZONED REZONES RADII RICIN
RIGID RISHI SIGIL TEIID TIBIA TIKIS TIMID TIPIS TITIS TORII VIGIL VILLI VINIC
VIRID VISIT VIVID VIZIR ZITIS ZIZIT

5

4 Z

5 Y

2 A

6 G

7 K

Judges Output to Screen

NONE

JAGGY
LEGGY
LOGGY
MOGGY
MUGGY
NAGGY
RAGGY
SAGGY
SOGGY
VUGGY
WIGGY

AA

NONE

OUTTALK
OUTTASK

12. Word Search

Program Name: WordSearch.java

Input File: wordsearch.dat

Judges Input File

18

```
O B E A Q U I L A M W E R D N A Q I
B A C D O B E N J A M I N O S S S S
E L I Z A B E T H I B A O R U I U A
H A N T E H A I R R T E D C T M R U
P A U Z P A T D I I U A H A I E A L
R M E H P H B R I M A Q B S T O Z H
I J U L O C I R A A I J E I H N A A
S T Y P P A L L A M H N O M T T L K
C D H L H R O E E H O O E S A H A E
I E T U I D I S H M A E L H E R A B
L B O Z L A S Q N A O M I N P P Y E
L O M A I H G A B E D N I G O E H R
A R I B P S L I K H A L I L E D T O
H A T D N R E U B E N E L I J A H S
S H P I S A I A H A S O L O M O N I
I M O L E U N E P N S I U R E U E L
L E H C A R A E I N H S A V Y N V A
E S T H E R D J O H N E H U M D E S
```

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```
ABEDNIGO
ABIGAIL
ABRAHAM
ADAM
AHITHOPHEL
ANDREW
APOLLOS
AQUILA
BALAAM
BATHSHEBA
BENJAMIN
DANIEL
DEBORAH
DELILAH
DORCAS
ELIJAH
ELISHA
ELIZABETH
ESTHER
EUNICE
EVE
ISAIHA
ISHMAEL
JEZEBEL
JOB
JOHN
JOSEPH
LAZARUS
LEAH
LOIS
```

(continued on next page)

12. Word Search (cont.)

MARTHA
MARY
MIRIAM
NAOMI
NOAH
OBADIAH
ONESIMUS
PENUEL
PHILEMON
PHILIP
PHUT
PRISCILLA
RACHEL
REBEKAH
REUBEN
REUEL
SAMSON
SAUL
SILAS
SIMEON
SOLOMON
STEPHEN
TABITHA
TIMOTHY
TITUS
VASHNI
ZABDI

Judges Output to Screen

ABEDNIGO 12 8
ABIGAIL 15 10
ABRAHAM 5 6
ADAM 5 14
AHITHOPHEL 2 10
ANDREW 1 16
APOLLOS 10 17
AQUILA 1 4
BALAAM 1 2
BATHSHEBA 10 18
BENJAMIN 2 6
DANIEL 18 7
DEBORAH 9 2
DELILAH 13 16
DORCAS 1 14
ELIJAH 14 12
ELISHA 18 1
ELIZABETH 3 1
ESTHER 18 1
EUNICE 6 3
EVE 16 17
ISAIAH 15 4
ISHMAEL 10 7
JEZEBEL 7 2

(continued on next page)

12. Word Search (cont.)

JOB 0 0
JOHN 18 8
JOSEPH 7 12
LAZARUS 8 17
LEAH 7 4
LOIS 8 7
MARTHA 8 10
MARY 8 14
MIRIAM 6 10
NAOMI 11 9
NOAH 0 0
OBADIAH 2 5
ONESIMUS 9 11
PENUEL 16 9
PHILEMON 5 5
PHILIP 8 5
PHUT 5 5
PRISCILLA 5 1
RACHEL 17 6
REBEKAH 12 18
REUBEN 14 6
REUEL 16 14
SAMSON 0 0
SAUL 2 18
SILAS 14 18
SIMEON 2 16
SOLOMON 15 11
STEPHEN 14 18
TABITHA 4 11
TIMOTHY 14 3
TITUS 6 15
VASHNI 17 14
ZABDI 11 4